

Problems in Poisson Distribution

TEST YOUR KNOWLEDGE

- Fit a Poisson distribution to the following :

x :	0	1	2	3	4
f :	192	100	24	3	1
 - If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals more than two will get a bad reaction.
 - If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the standard deviation.
 - If a random variable has a Poisson distribution such that $P(1) = P(2)$, find
 (i) mean of the distribution (ii) $P(4)$.
 - Suppose that X has a Poisson distribution. If $P(X = 2) = \frac{2}{3} P(X = 1)$ find, (i) $P(X = 0)$ (ii) $P(X = 3)$.
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- A certain screw making machine produces on average 2 defective screws out of 100, and packs them in boxes of 500. Find the probability that a box contains 15 defective screws.
 - The incidence of occupational disease in an industry is such that the workmen have a 10% chance of suffering from it. What is the probability that in a group of 7, five or more will suffer from it ?
 - (i) Fit a Poisson distribution to the following and calculate theoretical frequencies :

x :	0	1	2	3	4
f :	122	60	15	2	1

 (ii) Fit a Poisson distribution to the following data given the number of yeast cells per square for 400 squares :

<i>No. of cells per sq.</i> :	0	1	2	3	4	5	6	7	8	9	10
<i>No. of squares</i> :	103	143	98	42	8	4	2	0	0	0	0
 - Suppose a book of 585 pages contains 43 typographical errors. If these errors are randomly distributed throughout the book, what is the probability that 10 pages, selected at random, will be free from errors ?
 - Show that in a Poisson distribution with unit mean, mean deviation about mean is $\left(\frac{2}{e}\right)$ times the standard deviation.
 - In a certain factory turning razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 10000 packets.
 - The probability that a man aged 35 years will die before reaching the age of 40 years may be taken as 0.018. Out of a group of 400 men, now aged 35 years, what is the probability that 2 men will die within the next 5 years ?

Answers

- $320 \times \frac{e^{0.503} (9.503)^r}{r!}$
- 0.32
- 1
- (i) 2 (ii) $\frac{2}{3e^2}$
- (i) e^{-4} (ii) $4e^{-4}$
- $\frac{(10)^{15} e^{-10}}{(15)!} = 0.035$
- 0.0008
- (i) $121.36 \times \frac{(0.5)^r}{r!}$, where $r = 0, 1, 2, 3, 4$
 Theoretical frequencies are 121, 61, 15, 3, 0 respectively
 (ii) Theoretical frequencies are 109, 142, 92, 40, 13, 3, 1, 0, 0, 0
 9. 0.4795. 11. 9802, 196, 2
 12. 0.01936